

Mateusz Nasewicz

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Education

Wrocław University of Science and Technology — MSc in Applied Computer Science July 2026

Wrocław University of Science and Technology — BSc in Algorithmic Computer Science January 2025

Skills

Programming Languages: Java, SQL, TypeScript

Frameworks: Spring (Boot), Hibernate, Angular

Databases & Messaging: PostgreSQL, MongoDB, Redis, RabbitMQ, Kafka

Architecture & Web: REST API, Microservices, GraphQL, Websockets, OpenAPI

Testing: JUnit, Mockito, Testcontainers, Postman, Jasmine, Karma

Tools & Cloud: Git, Maven, Docker, AWS, Kubernetes, Terraform, Jenkins, Linux/Bash

Experience

Java Intern, RBO – Wrocław June 2025 – Sept 2025

- Developed an internal server monitoring tool using **Java and Spring Boot** to automate log parsing from virtual infrastructure (Hyper-V, VMware, KVM) and schedule daily status reports.
- Utilized **Hibernate** and **PostgreSQL** to persist parsed server metrics, ensuring efficient data retrieval for reports.
- Documented internal **REST APIs** using **OpenAPI (Swagger)** and maintained technical integration guides within **Confluence**.

Projects

Dopamine Delivery - Live 3D Tracking [DEPLOYED] | *Java, Spring Boot, Terraform, Angular, MongoDB, Stripe*

- Developed a responsive **Angular** frontend featuring high-performance, **live 3D tracking** of vehicle movements driven by **real-time data streaming** via secured **WebSockets (WSS)**.
- Integrated multiple third-party **REST APIs**, including **Stripe** for payments, **OSRM** for dynamic routing, and **Komoot Photon** for highly efficient geospatial address geocoding.
- Engineered an efficient backend geospatial search using **MongoDB** to process and query **OpenStreetMap (OSM)** data, rapidly retrieving nearby restaurants based on user coordinates.
- Provisioned infrastructure on **DigitalOcean** using **Terraform (IaC)**, deploying via automated **GitHub Actions CI/CD** pipelines with **Nginx** as a secure reverse proxy.

Urban Bicycle Infrastructure Management System | *Java, Spring Boot, PostgreSQL, Kafka, MQTT, Redis, AWS*

- Co-designed a scalable **microservices architecture** with domain separation (**DDD**) and multi-tier secure deployments on **AWS** for over 10,000 active IoT devices.
- Implemented real-time **IoT telemetry ingestion** via **MQTT (Mosquitto)**, caching live vehicle coordinate streams in **Redis** for low-latency geospatial queries.
- Ensured distributed transaction integrity across microservices by implementing the **Saga orchestration pattern** and high-throughput **Apache Kafka** event streaming for secure rentals and billing workflows.

Culinary Recipe Manager | *Java, Spring Boot, RabbitMQ, Angular, PostgreSQL*

- Centralized microservice routing using **Spring Cloud Gateway** and secured APIs with **Spring Security (JWT)**.
- Implemented **CQRS** and asynchronous **RabbitMQ** messaging to optimize performance and scalability.
- Applied full-stack testing best practices using **JUnit, Mockito, Testcontainers, Jasmine, and Karma**.

Publications

[IEEE] Privacy-Preserving Real-Time Gesture Recognition Using Cloud-Trained Neural Networks